

PROTOCOL MANUAL

ACCESS LINK

SHAL-1000

Revision 0.01

Revision History

Revision	Date	History
0.01	2025-11-25	Initialize Draft

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1. Notation

1.1 Notation

In this document, numeric values are represented as follows:

Hexadecimal - 0x00, 0x01, 0x02 ... 0xFF

Decimal - 0, 1, 2, ... 255

Binary - 00000000b, 00000001b, 00000010b ... 11111111b

1.2 Data Types (Character Set)

The data types and character sets used in this document are as follows:

Ascii 9 - Ascii Code; data consists of numeric characters only.

Ascii X - Ascii Code; data may consist of any characters.

HexStr - Ascii Code; two characters represent one byte of hexadecimal value.

Hex - Hexadecimal Code

BCD - Binary Coded Decimal Code

2. Packet Structure

2.1 Command Packet

	Header area			Data area	Trailer area
Field	STX	Length	Command	Data ¹⁾	ETX
Byte(s)	1	1	1	Variable	1

2.2 Response Packet

	Header area			Data area	Trailer area
Field	STX	Length	Command	Data ¹⁾	ETX
Byte(s)	1	1	1	Variable	1

¹⁾ If there is no data, the Data field is omitted.

2.3 Field Description

Field	Description	Remark
STX	Abbreviation for Start of Text. Indicates the start of a packet.	0x02
Length(1)	Indicates the total length of the packet. If there is no Data field, the length has the minimum value of 4.	
Command	Indicates the command to be executed by the device.	
Data	Indicates the data associated with the command.	
ETX	Abbreviation for End of Text. Indicates the end of a packet.	0x03

3. Error Code List

Code	Name	Description
0x00	COMM_SUCCESS	The command has been successfully executed.

4. Command Code List

4.1 Application Commands

Code	Name	BroadCast	Description
0x00	SET_RELAYCONTROL		Controls the relay.
0x01	SET_SERIALSEND		Outputs data via serial communication.
0x02	SET_WIEGANDOUT		Outputs data in Wiegand format.
0x09	GET_WIEGANDINPUTDATA		Transmits data received from Wiegand input.
0x10	GET_RECVDATA0		Transmits data received from RS-232C.
0x11	GET_RECVDATA1		Transmits data received from RS-485.
0x12	GET_INPUTPORT0		Transmits when Input Port 0 is detected as pressed.
0x13	GET_INPUTPORT1		Transmits when Input Port 1 is detected as pressed.

5. Command Description

SET_RELAYCONTROL (0x00) SET

Command Packet				Response Packet	
Command	0x00			Command	
Type	SET			Type	
DataSize	4			DataSize	
Data	Control			Data	
Control (Size:4)					
Name	Type	Size	Range	Description	
UseRelay	Ascii 9	1	0 ~ 2	Indicates the relay number. 0 - Relay 0 + Relay 1 1 - Relay 0 2 - Relay 1	
OutputType	Ascii 9	1	0 ~ 1	ndicates the output status. 0 - OFF 1 - ON	
Time	Ascii 9	2	All	Relay output time. If set to 0, the relay remains continuously output. This field is not applicable when the relay is OFF.	
This command is used to control the relay.					

SET_SERIALSEND (0x01) SET

Command Packet		Response Packet		
Command	0x01	Command		
Type	SET	Type		
DataSize	2 ~ 65	DataSize		
Data	SerialData	Data		
SerialData (Size: 2 ~ 65)				
Name	Type	Size	Range	Description
UseSerialPort	Ascii 9	1	0 ~ 1	Indicates the port to output data. 0 - RS-232C 1 - RS-485
SendData	Hex	1 ~ 64	All	Indicates the data to be output.
This command is used to send data via serial communication.				

SET_WIEGANDOUT (0x02) SET

Command Packet		Response Packet	
Command	0x02	Command	
Type	SET	Type	
DataSize	2 ~ 17	DataSize	
Data	WiegandData	Data	

Format (Size: 2 ~ 17)

Name	Type	Size	Range	Description
UseParity	Ascii 9	1	0 ~ 1	Indicates whether to output parity bits. 0 - Output 1 - Do not output
WiegandData	Hex	1 ~ 16	All	Indicates the data to be output.

This command is used to send data in Wiegand format.

GET_WIEGANDINPUTDATA (0x09) GET

Command Packet		Response Packet		
Command		Command	0x10	
Type		Type	GET	
DataSize		DataSize	8, 10	
Data		Data	RData	
RData (Size: 8, 10)				
Name	Type	Size	Range	Description
Data	Ascii 9	8, 10	All	Indicates the data received from Wiegand input. 8 bytes : 26-bit 10 bytes : 34-bit
This command is used to receive data transmitted from the Wiegand input.				

GET_RECVDATA0 (0x10) GET

Command Packet		Response Packet		
Command		Command	0x10	
Type		Type	GET	
DataSize		DataSize	1 ~ 64	
Data		Data	RData	
RData (Size: 1 ~ 64)				
Name	Type	Size	Range	Description
Data	Hex	1 ~ 64	All	Indicates the data received from RS-232C.
This command is used to receive data transmitted from RS-232C.				

GET_RECVDATA1 (0x11) GET

Command Packet		Response Packet		
Command		Command	0x11	
Type		Type	GET	
DataSize		DataSize	1 ~ 64	
Data		Data	RData	
RData (Size: 1 ~ 64)				
Name	Type	Size	Range	Description
Data	Hex	1 ~ 64	All	Indicates the data received from RS-485.
This command is used to receive data transmitted from RS-485.				

GET_INPUTPORT0 (0x12) GET

Command Packet		Response Packet		
Command		Command	0x12	
Type		Type	GET	
DataSize		DataSize	1	
Data		Data	Status	
Status (Size: 1)				
Name	Type	Size	Range	Description
Status	Hex	1	0 ~ 1	Indicates the status of Input Port 0. 0 : Released 1 : Pressed
This command is transmitted when the pressed or released status of Input Port 0 is detected.				

GET_INPUTPORT1 (0x13) GET

Command Packet		Response Packet		
Command		Command	0x13	
Type		Type	GET	
DataSize		DataSize	1	
Data		Data	Status	
Status (Size: 1)				
Name	Type	Size	Range	Description
Status	Hex	1	0 ~ 1	Indicates the status of Input Port 1. 0 : Released 1 : Pressed
This command is transmitted when the pressed or released status of Input Port 1 is detected.				